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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

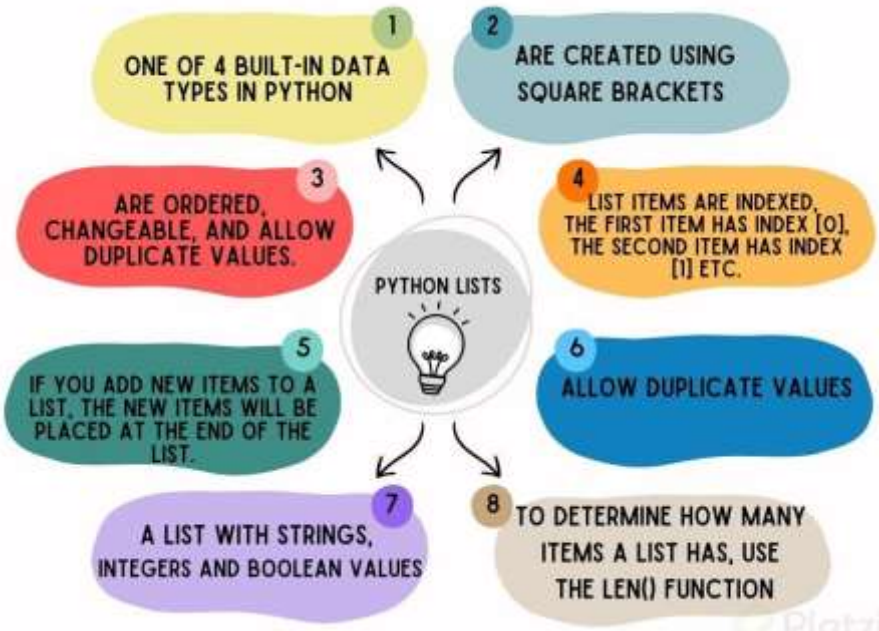
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 33: List Methods (Python)







Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a List is.
- Learn different List Methods in Python.
- Add, remove, and modify list items.
- Sort and reverse lists.
- Count and search elements.
- Build simple list-based programs.

What is a List?

A **List** is a collection of multiple items stored in a single variable.

Lists are written using **square brackets [ ]**.

Example:

fruits = ["Apple", "Banana", "Mango"]

A list can store:

- Fruits
- Books
- School items

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- Numbers
- Student names

Example

numbers = [10, 20, 30, 40]

colors = ["Red", "Blue", "Green"]

students = ["Rahul", "Asha", "Ali"]

Why Do We Use Lists?

Without a list:

fruit1 = "Apple"

fruit2 = "Banana"

fruit3 = "Mango"

With a list:

fruits = ["Apple", "Banana", "Mango"]

Easier to manage many values.

Common List Methods

Method	Purpose
append()	Add item at the end
insert()	Add item at a specific position
extend()	Add multiple items
remove()	Remove an item
pop()	Remove item by position
clear()	Remove all items
sort()	Arrange items in order

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Method	Purpose
reverse()	Reverse the list
count()	Count occurrences
index()	Find item position
copy()	Make a copy of the list
len()	Count total items

1️⃣ append()

Adds a new item to the end.

```
fruits = ["Apple", "Banana"]

fruits.append("Mango")

print(fruits)
```

Output

```
['Apple', 'Banana', 'Mango']
```

2️⃣ insert()

Adds an item at a specific position.

```
fruits = ["Apple", "Mango"]

fruits.insert(1, "Banana")

print(fruits)
```

Output

```
['Apple', 'Banana', 'Mango']
```

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3️⃣ extend()

Adds multiple items.

```
fruits = ["Apple"]

fruits.extend(["Banana", "Mango"])

print(fruits)
```

Output

```
['Apple', 'Banana', 'Mango']
```

4️⃣ remove()

Removes a specific item.

```
fruits = ["Apple", "Banana", "Mango"]

fruits.remove("Banana")

print(fruits)
```

Output

```
['Apple', 'Mango']
```

5️⃣ pop()

Removes an item using its position (index).

```
fruits = ["Apple", "Banana", "Mango"]

fruits.pop(1)

print(fruits)
```

Output

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['Apple', 'Mango']

6 clear()

Removes all items.

```
fruits = ["Apple", "Banana"]

fruits.clear()

print(fruits)
```

Output

```
[]
```

7 sort()

Sorts items in alphabetical or ascending order.

```
numbers = [50, 10, 30, 20]

numbers.sort()

print(numbers)
```

Output

```
[10, 20, 30, 50]
```

8 reverse()

Reverses the list.

```
numbers = [10, 20, 30]

numbers.reverse()
```

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print(numbers)

Output

[30, 20, 10]

9️⃣ count()

Counts how many times an item appears.

numbers = [1, 2, 2, 3, 2]

print(numbers.count(2))

Output

3

🔟 index()

Finds the position of an item.

fruits = ["Apple", "Banana", "Mango"]

print(fruits.index("Banana"))

Output

1

1️⃣1️⃣ copy()

Creates another list with the same items.

fruits = ["Apple", "Banana"]

new_list = fruits.copy()

print(new_list)

Output

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['Apple', 'Banana']

1️⃣2️⃣ len()

Counts total items.

fruits = ["Apple", "Banana", "Mango"]

print(len(fruits))

Output

3

Memory Trick

Method	Easy Meaning
append()	Add one item
insert()	Add at position
extend()	Add many items
remove()	Remove by value
pop()	Remove by position
clear()	Remove everything
sort()	Arrange in order
reverse()	Turn around
count()	Count items
index()	Find position
copy()	Make duplicate
len()	Total items

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Python List Methods

by levelupprogress

	Description	Code example	Diagram
append()	Adds an element to the end of the list	letters = ['a', 'b', 'c'] letters.append('d')	
extend()	Adds the elements of a list to the end of another list	letters = ['a', 'b', 'c'] letters.extend(['d', 'e', 'f'])	
insert()	Adds an element at a specific index	letters = ['a', 'b', 'c'] letters.insert(1, 'x')	
remove()	Removes the first occurrence of an element	letters = ['a', 'b', 'c', 'b'] letters.remove('b')	
pop()	Removes and returns the element at a given index	letters = ['a', 'b', 'c', 'd'] letter = letters.pop(1)	
count()	Returns the number of times a value appears in a list	letters = ['c', 'b', 'c', 'c', 'd'] print(letters.count('c'))	
sort()	Sorts the list in ascending order	letters = ['c', 'a', 'd', 'e', 'b'] letters.sort()	
reverse()	Reverses the order of elements in the list	letters = ['a', 'b', 'c'] letters.reverse()	

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APPEND
Add single element to end of list

```
office = ["Dwight", "Pam", "Michael"]
office.append("Jim")
```

CLEAR
Remove all items from list

```
office = ["Dwight", "Pam", "Michael"]
office.clear()
```

COPY
Return shallow copy of list

```
office = ["Dwight", "Pam", "Michael"]
office_copy = office.copy()
```

COUNT
Return count of an element in list

```
office = ["Dwight", "Pam", "Michael"]
count_pam = office.count("Pam")
```

EXTEND
Adds iterable elements to end of list

```
office = ["Dwight", "Pam", "Michael"]
office.extend(["Andy", "Jim"])
```

INDEX
Return index of first element in list matching given value

```
office = ["Dwight", "Pam", "Michael"]
index_of_Pam = office.index("Pam")
```

INSERT
Insert element to list at given index

```
office = ["Dwight", "Pam", "Michael"]
office.insert(1, "Andy")
```

POP
Remove element at given index and returns it

```
office = ["Dwight", "Pam", "Michael"]
office.pop(1)
```

REMOVE
Remove first item from list that has given value

```
office = ["Dwight", "Pam", "Michael"]
office.remove("Pam")
```

REVERSE
Reverse the list

```
office = ["Dwight", "Pam", "Michael"]
office.reverse()
```

SORT
Sort elements of a list

```
office = ["Dwight", "Pam", "Michael"]
office.sort()
```

Practical Session

Practical Exercise 1

Add a fruit.

```
fruits = ["Apple", "Banana"]


fruit = input("Enter a fruit: ")


fruits.append(fruit)


print(fruits)
```

Practical Exercise 2

Insert a student name.

```
students = ["Ali", "Ravi"]


students.insert(1, "Asha")


print(students)
```

Practical Exercise 3

Remove a fruit.

```
fruits = ["Apple", "Banana", "Mango"]


fruit = input("Enter fruit to remove: ")


fruits.remove(fruit)
```

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```
print(fruits)
```

Practical Exercise 4

Sort numbers.

```
numbers = [45, 10, 80, 25]


numbers.sort()


print(numbers)
```

Practical Exercise 5

Reverse a list.

```
colors = ["Red", "Green", "Blue"]


colors.reverse()


print(colors)
```

Practical Exercise 6

Count a number.

```
numbers = [10, 20, 20, 30, 20]


print(numbers.count(20))
```

Practical Exercise 7

Find a position.

```
animals = ["Cat", "Dog", "Rabbit"]
```

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print(animals.index("Dog"))

Practical Exercise 8

Copy a list.

list1 = ["Pen", "Book"]

list2 = list1.copy()

print(list2)

Practical Exercise 9

Commented [H1]:

Count total subjects.

subjects = ["Math", "Science", "English", "Computer"]

print(len(subjects))

Practical Exercise 10

Add multiple colors.

colors = ["Red"]

colors.extend(["Blue", "Green"])

print(colors)

Mini Project: Student Attendance List

students = []

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```
students.append(input("Enter Student 1: "))
students.append(input("Enter Student 2: "))
students.append(input("Enter Student 3: "))

print("\nStudent List")
print(students)

print("Total Students:", len(students))

students.sort()

print("Sorted List:", students)
```

Sample Output

```
Enter Student 1: Rahul
Enter Student 2: Asha
Enter Student 3: Ali

Student List
['Rahul', 'Asha', 'Ali']

Total Students: 3

Sorted List:
['Ali', 'Asha', 'Rahul']
```

Classroom Activity

Create a list of **5 favorite fruits**.

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Perform these operations:

- append()
- remove()
- insert()
- sort()
- reverse()

Observe the output after every step.

Practice Questions

1. What is a List?
2. Which brackets are used for lists?
3. Which method adds an item at the end?
4. Which method removes a specific item?
5. What is the use of sort()?
6. What does reverse() do?
7. Which method counts repeated items?
8. Which method finds an item's position?
9. What is the use of copy()?
10. Which function counts the total number of items?

Homework

Write a Python program to:

- Create a list of **5 subjects**.
- Add one more subject.
- Remove one subject.
- Sort the list.
- Reverse the list.
- Display the total number of subjects.

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Summary

- A **List** stores multiple values in one variable.
- Lists use **square brackets [ ]**.
- append() adds one item.
- insert() adds an item at a specific position.
- extend() adds multiple items.
- remove() deletes an item by value.
- pop() deletes an item by position.
- clear() removes all items.
- sort() arranges items.
- reverse() reverses the order.
- count() counts repeated items.
- index() finds an item's position.
- copy() creates a duplicate list.
- len() counts the total number of items.

Fun Challenge

Write a Python program that:

1. Creates a list of your **favorite games**.
2. Adds a new game using append().
3. Removes one game using remove().
4. Sorts the list alphabetically.
5. Displays:
    - The final list
    - The total number of games
    - The position of your favorite game using index()

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